

The map $\mathbf{u} \mapsto W(\mathbf{u})$, $\mathbf{u} \in \mathbb{C}^d$ is called the *Weyl representation* of the additive group $\mathbb{C}^d$. Now write $d = d_1 + d_2$, where $d_1, d_2 \in \mathbb{N}$. Then $\mathbb{C}^d = \mathbb{C}^{d_1} \oplus \mathbb{C}^{d_2}$, and the map given by $|e(\mathbf{v}_1 \oplus \mathbf{v}_2)\rangle \mapsto |e(\mathbf{v}_1)\rangle \otimes |e(\mathbf{v}_2)\rangle$, for all $\mathbf{v}_1 \in \mathbb{C}^{d_1}, \mathbf{v}_2 \in \mathbb{C}^{d_2}$ extends to a Hilbert space isomorphism between $\Gamma(\mathbb{C}^d)$ and $\Gamma(\mathbb{C}^{d_1}) \otimes \Gamma(\mathbb{C}^{d_2})$. Under this isomorphism, we identify these two Fock spaces and write $\Gamma(\mathbb{C}^d) = \Gamma(\mathbb{C}^{d_1}) \otimes \Gamma(\mathbb{C}^{d_2})$ and we have $W(\mathbf{u}_1 \oplus \mathbf{u}_2) = W(\mathbf{u}_1) \otimes W(\mathbf{u}_2)$, for all $\mathbf{u}_1 \in \mathbb{C}^{d_1}, \mathbf{u}_2 \in \mathbb{C}^{d_2}$ called the *factorizability* of the Weyl representation. Finally, the map $\mathbf{u} \mapsto W(\mathbf{u})$, $\mathbf{u} \in \mathbb{C}^d$ is a strongly continuous, projective, factorizable and irreducible unitary representation of the additive abelian group $\mathbb{C}^d$.

Another important class of unitary operators we need in this article consists of the so called *Bogoliubov transformations* on $\Gamma(\mathbb{C}^d)$. Given a symplectic transformation L on $\mathbb{C}^d$, there exists a unique unitary $\Gamma(L)$ on $\Gamma(\mathbb{C}^d)$, called the *Bogoliubov unitary* at L , satisfying $\Gamma(L)W(\mathbf{u})\Gamma(L)^\dagger = W(L\mathbf{u})$, and $\langle e(0)|\Gamma(L)|e(0)\rangle > 0$. In this case, we have $\Gamma(L)^\dagger = \Gamma(L^{-1})$. We refer the reader to [Par92, Chapter 2] and [JP21, Section IV] for more details on the topics mentioned in this sub section.

2.3. Gaussian States. In this section, we define Gaussian states and describe their properties which we need in a basis independent way. This will help us provide rigorous proofs later.

Definition 2.1. Let ρ be a state on $\Gamma(\mathbb{C}^d)$. Then the map $\hat{\rho} : \mathbb{C}^d \rightarrow \mathbb{C}$ defined by

(3)

is called the *quantum characteristic function* of ρ . If there exist a vector $\mathbf{m} \in \mathbb{C}^d$ and a real linear operator S on $\mathbb{C}^d$ such that

$$\hat{\rho}(\mathbf{z}) = \exp\left\{-i \operatorname{Re} \langle \mathbf{m} | \mathbf{z} \rangle - \frac{1}{2} \operatorname{Re} \langle \mathbf{z} | S | \mathbf{z} \rangle\right\}, \quad \forall \mathbf{z} \in \mathbb{C}^d, \quad (4)$$

then ρ is called a *d-mode quantum Gaussian state* or simply a *Gaussian state*. In this case, the pair $(\mathbf{m}, S)$ is unique; the vector $\mathbf{m}$ is called the mean vector and S is called the covariance operator of the Gaussian state ρ ; and we write $\rho = \rho_{(\mathbf{m}, S)}$.

Let ρ be a Gaussian state with mean vector $\mathbf{m}$ and covariance operator S . If V denotes an identification of $\mathbb{C}^d$ with $\mathbb{R}^{2d}$ as described in Section 2, then the quantum characteristic function of ρ can be considered as the map $\hat{\rho} : \mathbb{R}^{2d} \rightarrow \mathbb{C}$ given by

$$\hat{\rho}(V\mathbf{z}) = \exp\left\{-i(V\mathbf{m})^T(V\mathbf{z}) - \frac{1}{2}(V\mathbf{z})^T(VSV^T)(V\mathbf{z})\right\}, \quad \forall \mathbf{z} \in \mathbb{C}^d, \quad (5)$$

where $VSV^T \in M_{2d}(\mathbb{R})$. When the identification is clear from the context, we abuse the notation and write $\mathbf{m} \in \mathbb{R}^{2d}$ and $S \in M_{2d}(\mathbb{R})$. Furthermore, an arbitrary matrix $S \in M_{2d}(\mathbb{R})$ is the covariance matrix of a Gaussian state if and only if it belongs to the set

$$\mathcal{CM}(d) := \{A \in M_{2d}(\mathbb{R}) : A + iJ \geq 0\},$$

where J is taken as J_{2d} or $J_{2d_1} \oplus J_{2d_2}$ according to the identification we have chosen. For more details on quantum Gaussian states, we refer to [Par10], and for the basis-independent description of Gaussian states, see [Joh18]. The following proposition is a consequence of the definitions.